Database constraints improve your data model but there are operational challenges to adding them to production databases. You'll learn about those challenges and how to solve them.

**Correctness Terminology**



- Referential Integrity — Ensuring rows in two tables with common values both exist
- Constraints — Database object used to constrain data in a field or between tables
- Enum — Database list object to constrain possible field values
- Domain — Like an enum but can be shared across tables
- Correctness — Ensuring that data is consistent with the constraints

PostgreSQL provides a variety of constraint types. They are listed below and will be covered in greater detail as the chapter unfolds.

- PRIMARY KEY
- NOT NULL
- UNIQUE (covered in on page 60)
- REFERENCES (Foreign Key)
- CHECK
- EXCLUDE (Exclusion)

First up is the UNIQUE constraint.

## Multiple Column Uniqueness

Understanding which fields or attributes should be unique is one of the interesting challenges in data modeling.

Some tables start out with one field defining uniqueness, but then the definition changes over time. For example a City name might be unique for a single state, but exist as a name in many states when all States in the country are considered. Within the United States a city named Portland could refer to Portland, Maine or Portland, Oregon. Another example is Hollywood, which probably refers to Los Angeles but could refer to Florida. To express a unique city name you might want to define uniqueness as the combination of city name and state name.

In Rideshare there is a locations table with city and state columns. The table has a UNIQUE constraint on the address column.